Assignment – Week 1

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Preliminary remarks

You are expected to complete **two tasks** for this week, ideally of different types (e.g., one essay, one simulation, one formal derivation) – though this is not a strict requirement. The deadline is midnight on the 5th of May. You are strongly encouraged to explore additional tasks – this is an opportunity to deepen your understanding by tackling what most interests or challenges you.

The rationale behind offering many different tasks is twofold: to give you maximum choice, and to help you develop a sense of what it means to examine models carefully. Whether through formal derivation, coding, or conceptual writing, the aim of these assignments is to stretch yourselves intellectually. Try things, fail at them, and then try again. **Hitting walls is not only normal – it is essential when working with formal models and doing research more generally.** Much of the value lies in grappling with structure, assumptions, and logic – not just in arriving at the "correct" answer.

Please feel free to choose the level of difficulty that suits you best. Some tasks are more involved than others, and it is entirely fine to focus on those you find most manageable or intriguing. **There is absolutely no reason to feel bad about choosing easier options** – this material is tricky, and any serious, good-faith effort is genuinely appreciated.

This is a formative course – your marks do not count, so this is also a chance to take risks and venture out of your comfort zone. If time is tight and you cannot get to some of the more challenging exercises, that is completely fine. Do what you can, and feel free to keep the sheet for future reference.

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The use of generative AI is actively encouraged. It can help illuminate reasoning gaps, accelerate coding, or suggest alternative formulations. That said, a word of caution: outsourcing your thinking too early or too often can undermine your learning. **The long-run payoff comes from engaging deeply first, and only then consulting AI to refine or test your ideas.** I will not police the use of AI – the responsibility for learning is ultimately yours.

For simulation tasks, please include enough information for me to understand what you did: the parameter values, the random seed if you used one, the main lines of code or pseudocode, the plot or table, and a short interpretation. If your simulated result is meant to match an analytic benchmark, report both.

Submission instructions: Please email your completed tasks to me by the deadline. Ideally, submit your work as a single integrated document – such as an R Markdown or Jupyter Notebook file – combining code, output, and explanatory text. This format is especially useful for simulations and makes your reasoning easier to follow. You may also submit essays in Word, LaTeX, or plain Markdown. Although zipped folders are acceptable (for example, if you submit multiple files or raw code), they are not encouraged.

A stylised model of self-enforcing democracy

Reminder. *The loser receives L out of office and R in office, so the stakes of power are $S \equiv R - L > 0$. If the loser accepts defeat, she returns to office with probability π. If she fights, she wins with probability p and pays a one-time cost C. Let $\kappa \equiv (1-\delta)C$ be the per-period equivalent of that cost (Przeworski, 1991; Przeworski et al., 2015; Fearon, 2011).*

1. Compliance, costs, and stakes. *(formal)*

(a) Starting from

$$V_E = L + \frac{\pi S}{1-\delta}, \qquad V_F = L + \frac{pS}{1-\delta} - C,$$

derive the compliance condition $\pi S \geq pS - \kappa$ and the minimum return probability $\pi_{\min}(S) = p - \kappa/S$.

(b) Now suppose the cost of fighting is not one-time, but a recurring per-period cost c. Derive the new compliance condition. In one or two sentences, explain why the one-time and recurring-cost cases differ.

(c) Suppose the probability of returning to office falls as stakes rise: $\pi(S) = \bar{\pi} - \gamma S$, with $\gamma \geq 0$. Find the condition on S for compliance. If $\gamma > 0$, solve for the maximum sustainable stakes $S_{\max}$. How does this compare to the slide case where $\gamma = 0$?

(d) Interpret the result. Why does raising the stakes of office make democracy harder to sustain here, even though higher stakes will make reelection incentives stronger in Ferejohn?

2. Visualising self-enforcement. *(simulation)*

Hint. *In R, create a vector, such as* `S <- seq(0.1, 2, length.out = 500)`*; in Python, use* `S = np.linspace(0.1, 2, 500)`*. Store each parameter combination in a small table/data frame before plotting. This makes it easier to facet or colour lines by parameter value.*

(a) Create a grid of stakes $S \in [0.1, 2]$. Using $(\delta, p, C) = (0.90, 0.60, 1.00)$, compute $\kappa = (1 - \delta)C$ and $\pi_{\min}(S) = p - \kappa/S$ at every grid point.

(b) Plot $\pi_{\min}(S)$ and add a horizontal line at $\bar{\pi} = 0.40$. Mark or shade the values of S for which $\pi_{\min}(S) \leq \bar{\pi}$. Check that the numerical cutoff matches the algebraic cutoff $S_{\max} = \kappa/(p - \bar{\pi})$.

(c) Vary one primitive at a time. Produce two additional panels: one comparing $C \in \{0.5, 1, 1.5\}$, and one comparing $p \in \{0.4, 0.6, 0.8\}$, holding the other parameters fixed. Which primitive shifts the self-enforcing range more in your plot?

(d) Now let the probability of returning to office fall with the stakes of the election: $\pi(S) = \bar{\pi} - \gamma S$ for $\gamma \in \{0, 0.1, 0.2\}$. Plot $\pi_{\min}(S)$ and $\pi(S)$ on the same graph. Where do they cross? How does this change the interpretation of "high stakes"?

(e) Briefly connect the plot back to real-world institutions. Which intervention expands the compliance region: lowering S, raising κ, lowering p, or raising $\bar{\pi}$? Give one real-world institutional example of each.

3. Empirical illustration (max. 700 words). *(essay)*

Choose one real-world example to illustrate the comparative statics from the self-enforcing democracy model. You may use either a cross-country comparison or a within-country, over-time comparison.

Your goal is to map the model's parameters to concrete political conditions. For example:

- What raises or lowers the stakes of office S? Think about patronage, control of the state, resource rents, constitutional limits, prosecution after leaving office, or threats to core groups.
- What makes fighting costly, i.e. what corresponds to κ? Think about sanctions, military fragmentation, repression costs, civil disorder, international pressure, or legal accountability.
- What shapes p, the probability of success by force? Think about control of the military, armed supporters, foreign backing, or organisational capacity.
- What makes future alternation credible, i.e. what shapes π? Think about electoral institutions, party-system openness, courts, federalism, term limits, or past experience with peaceful alternation.

Then explain how these parameters differ across your cases. Which parameter does most of the explanatory work? Did the likelihood of peaceful power transfer increase or decrease, and why?

4. **Where does $\pi(S)$ come from?** (max. 1000 words). *(essay / challenge)*

   In the model above, π is treated as given or exogenous. In reality, the probability that losers return to office through elections may itself depend on the stakes of office. Write an essay explaining why $\pi(S)$ might be flat, increasing, decreasing, or hump-shaped.

   A useful answer should identify a mechanism. For instance, higher stakes might increase mobilisation and therefore raise the opposition's chance of winning; but very high stakes might also encourage repression, fraud, opposition demobilisation, or institutional manipulation, lowering π. Your task is not to solve a full model, but to make the actor-level logic clear: who responds to higher stakes, what do they do, and why does this change the probability of future alternation?

   Optional formal sketch: Specify players, actions, payoffs, and timing for a simple model that could generate your proposed shape of $\pi(S)$.

Analysing the Ferejohn (1986) model

Reminder. *The incumbent privately observes a shock $\theta \sim F$ and chooses effort $a \geq 0$. Voters observe only performance $u = a\theta$ and reelect iff $u \geq K$. Let $\Delta \equiv \delta(V^I - V^O)$ be the discounted value of staying in office rather than losing.*

1. **Threshold-and-shirk, with a twist.** *(formal)*

(a) Show that, after observing θ, the incumbent either chooses $a = K/\theta$ and clears the cutoff, or chooses $a = 0$ and loses. Show that clearing is optimal iff $\phi(K/\theta) \leq \Delta$.

(b) With $F = \text{Uniform}[0, 1]$ and $\phi(a) = a$, derive the result:

$$K^* = \Delta/2, \qquad \theta^* = 1/2, \qquad \mathbb{E}[u] = \Delta/4.$$

Explain why the voter optimally tolerates shirking in half of all periods.

(c) Now change only the cost function to $\phi(a) = a^2/2$. Derive θ^*, the voter's expected performance $\mathbb{E}[u]$, and the optimal cutoff K^*. Does the incumbent still shirk half the time? What changes?

(d) Suppose the value of office W rises, widening Δ. Why does this strengthen accountability in Ferejohn but weaken self-enforcing democracy in the previous section?

2. Simulating electoral control. *(simulation)*

The aim is to see the threshold-and-shirk logic in simulated data. Your code should make it possible to compare effort, performance, and reelection across cost functions.

Hint. *Write one function that takes a vector of shocks, a cutoff K, a cost function $\phi(\cdot)$, and Δ, and returns effort, performance, and a reelection indicator. Then call the same function for the linear and quadratic cases. Avoid writing two unrelated simulations.*

(a) Set $\Delta = 1$ and simulate $T = 1000$ shocks $\theta_t \sim \text{Uniform}[0, 1]$. For the linear-cost case, use $K^* = \Delta/2$. In each period, apply the rule: if $\phi(K^*/\theta_t) \leq \Delta$, set $a_t = K^*/\theta_t$ and $u_t = K^*$; otherwise set $a_t = 0$ and $u_t = 0$.

(b) Check your simulation against the analytic benchmark: the reelection rate should be close to 1/2, and average performance should be close to $\Delta/4$.

(c) Repeat for the quadratic-cost case from part 1(c), using the K^* you derived there. Again record a_t, u_t, and reelection.

(d) Plot two things for both cost functions: the distribution of effort a_t and the distribution of observed performance u_t. Report the shirking rate, average effort conditional on not shirking, and average performance.

(e) Increase Δ from 1 to 2 and then lower it to 0.5. What changes mechanically? What changes substantively? Interpret Δ as the punishment for losing office.

(f) Interpret the whole exercise. What does the simulation teach you about the trade-off between asking for more performance when conditions are good and inducing zero effort when conditions are bad?

3. Are stationary cutoff rules suboptimal? (max. 1000 words). *(essay)*

Ferejohn's voter uses a stationary cutoff: the same performance threshold is applied every period. Write an essay engaging with at least two of the following points:

(a) In your own words, explain why stationary cutoff rules may be too rigid. Focus on the fact that the voter observes performance, not effort, and that the informational environment may change over time.

(b) Would a declining, increasing, or state-responsive cutoff improve accountability? Give one setting in which it would, and one in which it might make accountability worse.

(c) Incumbents are empirically more likely to be reelected – the incumbency advantage. Can Ferejohn-style models provide a rational explanation for this pattern? How does this compare to explanations based on voter bias, partisanship, or name recognition?

(d) Use one real-world example of retrospective voting. Identify the hidden action, the noisy signal, the cutoff voters appear to apply, and the punishment available to them.

Note: You are not expected to do formal modelling here. The aim is to reason carefully about when a simple cutoff is useful, when it is too crude, and what voters would need to know in order to improve on it.

Analysing the Fearon (1999) model

Reminder. *Fearon adds types. Good types choose the voter's ideal policy $x = 0$. Bad types prefer $x = 1$, but may moderate to improve reelection chances. Voters observe a noisy signal $z = -x^2 + \varepsilon$.*

1. Selection, sanctioning, and the commitment problem. *(formal)*

(a) **Pure selection.** Suppose good types choose $x = 0$ and bad types choose a fixed $x_b > 0$. With symmetric unimodal noise, show that the voter reelects iff $z \geq -x_b^2/2$. Why is the cutoff the midpoint between the two type means?

(b) **Pure sanctioning.** Suppose all politicians are bad types who choose x in period 1 and $x = 1$ in period 2. Their period-1 problem is

$$\max_x \; W - (1-x)^2 + \delta W \; \Pr(z \geq k).$$

Using $\Pr(z \geq k) = 1 - F(k + x^2)$, derive the first-order condition. Explain why the sanctioning-optimal cutoff sets $k = -x^2$, placing the density at its peak.

(c) Show that this gives $x^* = 1/[1 + \delta W f(0)]$. What happens to x^* when monitoring improves, i.e. when $f(0)$ rises?

(d) **Mixed case.** Now suppose there is a small probability $\alpha > 0$ of good types. The voter wants to use the selection cutoff $k = -x_b^2/2$ after seeing the signal. Plug this cutoff into the bad type's incentive problem. Why does the marginal incentive to moderate fall relative to pure sanctioning? Explain the commitment problem in words.

2. Simulating the selection–sanctioning trade-off. *(simulation)*

This exercise asks you to move from the two clean benchmark cases to the mixed case. Start with the benchmark, then add the complication.

Hint. *Use grid search. For example, create* `xgrid <- seq(0, 1, length.out = 1001)` *in R or* `xgrid = np.linspace(0, 1, 1001)` *in Python. For each candidate x, compute the payoff; the maximiser is the grid point with the largest payoff. This is approximate, but good enough here.*

(a) Fix $W = 1$, $\delta = 0.9$, and $\varepsilon \sim N(0, \sigma^2)$ with $\sigma = 0.5$. Compute the normal density at zero, $f(0)$, and use the pure-sanctioning formula

$$x^S = \frac{1}{1 + \delta W f(0)}, \qquad k^S = -(x^S)^2.$$

Interpret x^S: is the bad type close to the voter's ideal or close to its own ideal?

(b) Now take pure selection with a fixed bad-type choice $x_b = 0.5$. The selection cutoff is $k^P = -x_b^2/2$. Simulate 1000 good types ($x = 0$) and 1000 bad types ($x = x_b$). Report the reelection rate of each type and the share of reelected politicians who are bad.

(c) Approximate the mixed case. For each candidate $\hat{x} \in [0, 1]$, set the voter's selection cutoff to $k(\hat{x}) = -\hat{x}^2/2$. For that cutoff, find the bad type's best response $x^{BR}(\hat{x})$ by grid search over $x \in [0, 1]$:

$$W - (1 - x)^2 + \delta W \left[1 - \Phi\left(\frac{k(\hat{x}) + x^2}{\sigma}\right)\right].$$

Pick an approximate fixed point where $x^{BR}(\hat{x})$ is closest to $\hat{x}$. Compare this mixed-case x^M to x^S.

(d) Simulate 1000 good and 1000 bad politicians using x^M and the corresponding cutoff $k^M = -(x^M)^2/2$. Report the same quantities as in part (b).

(e) Repeat parts (a)–(d) for $\sigma = 0.25$ and $\sigma = 1$. What happens when signals become cleaner or noisier?

(f) In 3–4 sentences, connect the simulation to Fearon's claim that selection can undermine sanctioning. Where exactly does the incentive to moderate weaken?

3. Noise structures and voter mistakes. *(formal + simulation + substantive)*

Fearon's midpoint cutoff relies on a particular signal structure. This question asks you to examine this assumption.

Hint. *Use the same random seed and the same number of politicians in each case so that differences across panels are not just simulation noise. For false positives and false negatives, create two Boolean variables:* `bad_reelected` *and* `good_removed`*.*

(a) Start with the simple case: additive, symmetric, unimodal noise,

$$z = -x^2 + \varepsilon.$$

Explain formally why the good-type density is centred at 0, the bad-type density is centred at $-x_b^2$, and the selection cutoff is the midpoint $k = -x_b^2/2$.

(b) Now change only the variance of the additive normal noise. With $x_b = 0.5$, simulate 1000 good and 1000 bad types for $\sigma \in \{0.25, 0.5, 1\}$, applying the midpoint cutoff each time. Report false positives (bad types reelected) and false negatives (good types removed). What changes, and what does not?

(c) Now change the shape of the additive noise. Replace the normal shock with a mean-zero shifted lognormal shock. Keep using the midpoint cutoff. Does the cutoff still balance the two kinds of errors? Explain why skewness matters.

(d) Finally, consider multiplicative noise, for example $z = -x^2\varepsilon$ with $\varepsilon > 0$ lognormally distributed. Explain why the bad-type signal distribution is no longer just a translation of the good-type distribution. In your simulation, does a single cutoff in z still separate types well?

(e) Substantive interpretation: give one political setting where symmetric additive noise is plausible, one where skewed shocks are plausible, and one where multiplicative noise is plausible. How would each affect voters' ability to select good politicians?

4. Ignorance, sophistication, and voter commitment. *(formal + simulation + essay)*

This question develops an extension of Fearon's mixed case. The point is to reason through a tension: sophisticated voters are better at distinguishing effort from bad luck, but they may also be worse at credibly committing to a pure sanctioning rule.

Setup for parts (d)–(h). There are two voter types. A share β are sophisticated and update about politician type after observing the signal. A share $1-\beta$ are ignorant and, for simplicity, do not update about type. Given an anticipated bad-type choice x_b, ignorant voters use the sanctioning cutoff $k_I = -x_b^2$; sophisticated voters use the selection cutoff $k_S = -x_b^2/2$. The politician is reelected if a majority of voters vote to reelect. Assume $x_b > 0$, so $k_I < k_S$.

(a) **Start with Fearon.** Explain the commitment problem Fearon identifies. What cutoff would voters like to commit to ex ante if they wanted to discipline bad types? What cutoff do they want to use ex post once the signal also contains information about type?

(b) **Ignorance as commitment.** Explain why ignorance can restore the ability to commit to a sanctioning rule. Be precise: ignorant voters are not "better" because they know more; they are useful here because they do not condition their retention decision on type inference.

(c) **The two effects of sophistication.** As voters become more sophisticated, they may become better at distinguishing effort from bad luck. But they may also become more tempted to use the election for selection rather than sanctioning. State these two effects separately. Why do they point in different directions for electoral accountability?

(d) **Majority rule with two voter types.** Given $k_I < k_S$, divide the real line for signals z into three regions: $z < k_I$, $k_I \leq z < k_S$, and $z \geq k_S$. In each region, say how ignorant voters vote, how sophisticated voters vote, and whether the incumbent is reelected when $\beta < 1/2$ and when $\beta > 1/2$.

(e) **Effective cutoff.** Use part (d) to derive the effective electoral cutoff:

$$k_{\text{eff}}(\beta) = \begin{cases} k_I, & \beta < 1/2, \\ k_S, & \beta > 1/2, \end{cases}$$

ignoring the knife-edge case $\beta = 1/2$. What happens to the bad type's incentive to choose a low x_b when the electorate crosses from an ignorant majority to a sophisticated majority?

(f) **Simulation of the majority-rule extension.** Fix $W = 1$, $\delta = 0.9$, $\sigma = 0.5$, and a grid $\hat{x} \in [0, 1]$ of anticipated bad-type choices. Use

$$\beta \in \{0, 0.25, 0.49, 0.51, 0.75, 1\}.$$

For each value of β:

i. for each anticipated $\hat{x}$, compute $k_I(\hat{x}) = -\hat{x}^2$ and $k_S(\hat{x}) = -\hat{x}^2/2$;

ii. use part (e) to determine the effective cutoff $k_{\text{eff}}(\beta, \hat{x})$;

iii. for that cutoff, search over actual choices $x \in [0, 1]$ and compute the bad type's payoff

$$W - (1 - x)^2 + \delta W \left[1 - \Phi\left(\frac{k_{\text{eff}}(\beta, \hat{x}) + x^2}{\sigma}\right)\right];$$

iv. record the best response $x^{BR}(\beta, \hat{x})$;

v. find an approximate fixed point where $x^{BR}(\beta, \hat{x})$ is closest to $\hat{x}$. Call this $x_b^*(\beta)$.

Plot $x_b^*(\beta)$ against β. What discontinuity do you see around $\beta = 1/2$, and why?

(g) **Adding the benefit of sophistication.** Now allow sophistication to improve signal quality. Let the signal noise be $\sigma(\beta) = 1 - 0.5\beta$, so electorates with more sophisticated voters observe more informative signals. Repeat part (f). Does more sophistication now improve accountability, worsen it, or do both at different ranges of β?

(h) **Putting the pieces together.** Write a short interpretation. When are more sophisticated voters good for making elections "work"? When can sophistication undermine sanctioning? Your answer should explicitly weigh the informational benefit of sophistication against the commitment cost.

(i) **Push back.** Is the binary distinction between ignorant and sophisticated voters too crude? What would change if voters were partially sophisticated, if parties supplied cues, or if institutions could pre-commit voters to a performance metric?

5. **Selection versus sanctioning** (max. 900 words). *(essay)*

   Based on Fearon (1999), write a short essay addressing:

   (a) What is the difference between viewing elections as mechanisms of selection and sanctioning? Be explicit about what voters observe, what politicians choose, and what is fixed as a type.

   (b) Fearon argues that real-world voter behaviour often fits the selection view better than the pure sanctioning view. What kinds of empirical patterns motivate that claim?

   (c) In the mixed model, why can trying to select good types undermine incentives for bad types to behave well? Explain the commitment problem without simply repeating the algebra.

   (d) Do you think democratic elections work better as selection devices or sanctioning devices in practice? Pick one domain – corruption, competence, ideological extremism, economic management, candidate scandals – and reason through the trade-off.

6. **Are term limits good for accountability?** (max. 1000 words). *(essay)*

   In electoral models like Ferejohn's and Fearon's, reelection incentives help motivate good performance. Term limits remove that incentive after a fixed number of terms, but they may also help voters replace bad types when signals are noisy.

   Write an essay engaging with the following:

   (a) Why does the prospect of reelection create incentives for effort in sanctioning models? What happens in the final period of a politician's career?

   (b) If voters observe only noisy signals of politician quality, why might they fail to remove bad types even after one or two terms? How might term limits help in that case?

   (c) What are the costs of term limits? Consider effort, expertise, long-term projects, and the possibility of forcing out good types.

   (d) Overall, under what political conditions are term limits more likely to improve accountability, and under what conditions are they likely to damage it?

   (e) *Optional:* if voters had better audits, independent media, or policy-performance metrics, would the case for term limits become stronger or weaker?

References

Fearon, James D., "Electoral Accountability and the Control of Politicians: Selecting Good Types versus Sanctioning Poor Performance," in Adam Przeworski, Bernard Manin, and Susan C. Stokes, eds., *Democracy, Accountability, and Representation*, Cambridge University Press, 1999, pp. 55–97.

—, "Self-enforcing Democracy," *The Quarterly Journal of Economics*, 2011, 126 (4), 1661–1708.

Ferejohn, John, "Incumbent Performance and Electoral Control," *Public Choice*, 1986, 50 (1-3), 5–25. Accessed April 30, 2025.

Przeworski, Adam, *Democracy and the Market: Political and Economic Reforms in Eastern Europe and Latin America*, Cambridge University Press, 1991.

—, Gonzalo Rivero, and Tianyang Xi, "Elections as a Conflict Processing Mechanism," *European Journal of Political Economy*, 2015, 39, 235–248.